DBMS – SQL QUERY IMPLEMENTATION FOR E-COMMERCE DATABASE

Project Title:Myntra- E commerce database Management System skill

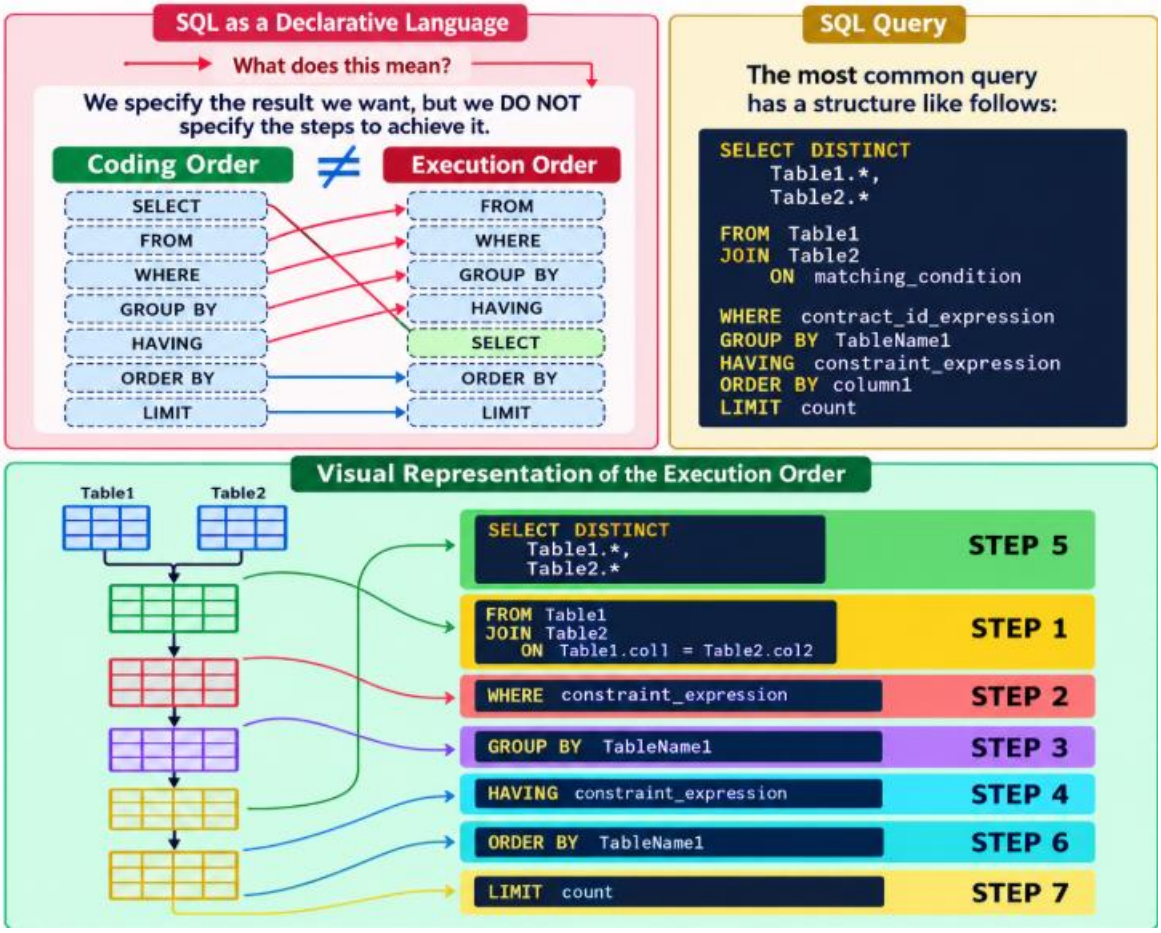
Objectives

1. Perform SELECT, WHERE, ORDER BY, and DISTINCT queries.
2. Search products based on price, category, and availability.
3. Retrieve customer and product information.
4. Apply filtering conditions.
5. Generate basic business reports.

1. Basic SQL Queries

1.1 SELECT Query

The SELECT statement is used to retrieve data from a table.



Display all products

```
SQL> SELECT * FROM Product;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID BRAND
----------- --------------------------------------------------
PRODUCT_SIZE         COLOR                          PRICE   DISCOUNT
-------------------- ------------------------------ ---------- ----------
     STOCK
----------
       201
Cotton Shirt
        101 Levis

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID BRAND
----------- --------------------------------------------------
PRODUCT_SIZE         COLOR                          PRICE   DISCOUNT
-------------------- ------------------------------ ---------- ----------
     STOCK
----------
M                    Blue                            1399         12
        50


PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID BRAND
----------- --------------------------------------------------
PRODUCT_SIZE         COLOR                          PRICE   DISCOUNT
-------------------- ------------------------------ ---------- ----------
     STOCK
----------
       202
Floral Dress
        102 Vero Moda

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID BRAND
----------- --------------------------------------------------
PRODUCT_SIZE         COLOR                          PRICE   DISCOUNT
-------------------- ------------------------------ ---------- ----------
     STOCK
----------
S                    Red                             1999         15
        30


PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID BRAND
----------- --------------------------------------------------
PRODUCT_SIZE         COLOR                          PRICE   DISCOUNT
-------------------- ------------------------------ ---------- ----------
```

Display selected product details

```
SQL> SELECT Product_ID, Product_Name, Brand, Price FROM Product;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
BRAND                                                    PRICE
-------------------------------------------------- ----------
       201
Cotton Shirt
Levis                                                    1399

       202
Floral Dress
Vero Moda                                                1999

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
BRAND                                                    PRICE
-------------------------------------------------- ----------

       203
Running Shoes
Nike                                                     3499

       204
Analog Watch

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
BRAND                                                    PRICE
-------------------------------------------------- ----------
Fossil                                                   5999

       205
Backpack
Wildcraft                                                1799
```

1.2 WHERE Clause

The WHERE clause is used to filter records based on a condition.

Find products with price less than 1000

```
SQL> SELECT * FROM Product WHERE Price < 2000;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID BRAND
----------- --------------------------------------------------
PRODUCT_SIZE         COLOR                          PRICE   DISCOUNT
-------------------- ------------------------------ ---------- ----------
     STOCK
----------
       201
Cotton Shirt
        101 Levis

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID BRAND
----------- --------------------------------------------------
PRODUCT_SIZE         COLOR                          PRICE   DISCOUNT
-------------------- ------------------------------ ---------- ----------
     STOCK
----------
M                    Blue                            1399         12
        50


PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID BRAND
----------- --------------------------------------------------
PRODUCT_SIZE         COLOR                          PRICE   DISCOUNT
-------------------- ------------------------------ ---------- ----------
     STOCK
----------
       202
Floral Dress
        102 Vero Moda
```

Find products with price greater than 1500

```
SQL> SELECT * FROM Product WHERE Price > 1500;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID BRAND
----------- --------------------------------------------------
PRODUCT_SIZE         COLOR                          PRICE   DISCOUNT
-------------------- ------------------------------ ---------- ----------
     STOCK
----------
       202
Floral Dress
        102 Vero Moda

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID BRAND
----------- --------------------------------------------------
PRODUCT_SIZE         COLOR                          PRICE   DISCOUNT
-------------------- ------------------------------ ---------- ----------
     STOCK
----------
S                    Red                             1999         15
        30


PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID BRAND
----------- --------------------------------------------------
PRODUCT_SIZE         COLOR                          PRICE   DISCOUNT
-------------------- ------------------------------ ---------- ----------
     STOCK
----------
       203
Running Shoes
        103 Nike

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID BRAND
----------- --------------------------------------------------
PRODUCT_SIZE         COLOR                          PRICE   DISCOUNT
-------------------- ------------------------------ ---------- ----------
     STOCK
----------
9                    Black                           3499         20
        40
```

1.3 ORDER BY

The ORDER BY clause is used to sort the query results.

Products from lowest to highest price

```
SQL> SELECT Product_ID, Product_Name, Price FROM Product ORDER BY Price ASC;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
       201
Cotton Shirt
      1399

       205
Backpack
      1799

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------

       202
Floral Dress
      1999

       203
Running Shoes

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
      3499

       204
Analog Watch
```

Products from highest to lowest price

```
SQL> SELECT Product_ID, Product_Name, Price FROM Product ORDER BY Price DESC;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
       204
Analog Watch
      5999

       203
Running Shoes
      3499

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------

       202
Floral Dress
      1999

       205
Backpack

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
      1799

       201
Cotton Shirt
```

1.4 DISTINCT

The DISTINCT keyword is used to display unique values.

Display unique categories

```
SQL> SELECT DISTINCT Category_ID FROM Product;

CATEGORY_ID
-----------
         101
         102
         103
         104
         105
```

Display unique brands

```
SQL> SELECT DISTINCT Brand FROM Product;

BRAND
----------------------------------------
Levis
Vero Moda
Nike
Fossil
Wildcraft
```

2. Search Products

2.1 Search Products Based on Price

```
SQL> SELECT Product_ID, Product_Name, Price FROM Product WHERE Price BETWEEN 1000 AND 4000;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
       201
Cotton Shirt
      1399

       202
Floral Dress
      1999

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------

       203
Running Shoes
      3499

       205
Backpack

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
      1799
```

2.2 Search Products Based on Category

```
SQL> SELECT Product_ID, Product_Name, Category_ID FROM Product WHERE Category_ID = 101;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID
-----------
       201
Cotton Shirt
```

2.3 Search Products Based on Availability

```
SQL> SELECT Product_ID, Product_Name, Stock FROM Product WHERE Stock > 30;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     STOCK
----------
       201
Cotton Shirt
        50

       203
Running Shoes
        40

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     STOCK
----------

       205
Backpack
        35
```

3. Retrieve Customer Information

Display all customers

```
SQL> SELECT Customer_ID, First_Name, Last_Name FROM Customer;

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME
--------------------------------------------------
       101 Rahul
Sharma

       102 Priya
Ravi

       103 Karthik
Kumar


CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME
--------------------------------------------------
       104 Anitha
Suresh

       105 Arun
Prakash

       106 Divya
Lakshmi


CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME
--------------------------------------------------
       107 Vignesh
Raj

       108 Meena
Krishnan
```

Search for a particular customer

```
SQL> SELECT * FROM Customer WHERE Customer_ID = 101;

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          GENDER
-------------------------------------------------- ----------
EMAIL
--------------------------------------------------------------------------------
PHONE
---------------
PASSWORD
--------------------------------------------------------------------------------
ADDRESS
--------------------------------------------------------------------------------
CITY
--------------------------------------------------
STATE                                                 PINCODE REGISTRAT
-------------------------------------------------- ---------- ---------
        101 Rahul

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          GENDER
-------------------------------------------------- ----------
EMAIL
--------------------------------------------------------------------------------
PHONE
---------------
PASSWORD
--------------------------------------------------------------------------------
ADDRESS
--------------------------------------------------------------------------------
CITY
--------------------------------------------------
STATE                                                 PINCODE REGISTRAT
-------------------------------------------------- ---------- ---------
Sharma                                             Male

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          GENDER
-------------------------------------------------- ----------
EMAIL
--------------------------------------------------------------------------------
PHONE
---------------
PASSWORD
--------------------------------------------------------------------------------
ADDRESS
--------------------------------------------------------------------------------
CITY
--------------------------------------------------
STATE                                                 PINCODE REGISTRAT
-------------------------------------------------- ---------- ---------
rahul@gmail.com
```

4. Apply Filtering Conditions

Multiple conditions can be combined using AND, OR, and NOT.

4.1 AND Condition

```
SQL> SELECT Product_ID, Product_Name, Price FROM Product
  2  WHERE Price < 3000 AND Category_ID = 102;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------
     PRICE
----------
       202
Floral Dress
      1999
```

4.2 OR Condition

```
SQL> SELECT Product_ID, Product_Name, Price FROM Product
  2  WHERE Category_ID = 101 OR Category_ID = 103;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------
     PRICE
----------
       201
Cotton Shirt
      1399

       203
Running Shoes
      3499

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------
     PRICE
----------
```

4.3 IN Operator

```
SQL> SELECT Product_ID, Product_Name, Price FROM Product
  2  WHERE Category_ID IN (101, 102, 103);

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
       201
Cotton Shirt
      1399

       202
Floral Dress
      1999

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------

       203
Running Shoes
      3499
```

4.4 BETWEEN Operator

```
SQL> SELECT Product_ID, Product_Name, Price FROM Product
  2  WHERE Price BETWEEN 1000 AND 2000;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------
     PRICE
----------
       201
Cotton Shirt
      1399

       202
Floral Dress
      1999

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------
     PRICE
----------

       205
Backpack
      1799
```

4.5 NOT Condition

```
SQL> SELECT Product_ID, Product_Name, Brand FROM Product WHERE NOT Brand = 'Nike';

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
BRAND
--------------------------------------------------
       201
Cotton Shirt
Levis

       202
Floral Dress
Vero Moda

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
BRAND
--------------------------------------------------

       204
Analog Watch
Fossil

       205
Backpack

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
BRAND
--------------------------------------------------
Wildcraft
```

5. Basic Business Reports

5.1 Product Price Report

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM Product
  3  ORDER BY Price DESC;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
       204
Analog Watch
      5999

       203
Running Shoes
      3499

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------

       202
Floral Dress
      1999

       205
Backpack

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
      1799

       201
Cotton Shirt
      1399
```

5.2 Customer Order Report

```
SQL> SELECT c.Customer_ID,
  2         c.First_Name || ' ' || c.Last_Name AS Customer_Name,
  3         o.Order_ID,
  4         p.Payment_Date,
  5         p.Amount
  6  FROM Customer c
  7  JOIN Orders o ON c.Customer_ID = o.Customer_ID
  8  JOIN Payment p ON o.Order_ID = p.Order_ID
  9  ORDER BY p.Payment_Date DESC;

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
  ORDER_ID PAYMENT_D     AMOUNT
---------- --------- ----------
        102
Priya Ravi
       502 16-SEP-26     3398.3

        106
Divya Lakshmi
       506 15-SEP-26    2462.24

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
  ORDER_ID PAYMENT_D     AMOUNT
---------- --------- ----------

        105
Arun Prakash
       505 14-SEP-26     1619.1

        104
Anitha Suresh

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
  ORDER_ID PAYMENT_D     AMOUNT
---------- --------- ----------
       504 13-SEP-26    5699.05

        103
Karthik Kumar
       503 12-SEP-26     2799.2

        101
```

5.3 Product Rating Report

```
SQL> SELECT p.Product_ID, p.Product_Name, AVG(r.Rating) AS Avg_Rating
  2  FROM Product p
  3  JOIN Rating r ON p.Product_ID = r.Product_ID
  4  GROUP BY p.Product_ID, p.Product_Name
  5  ORDER BY Avg_Rating DESC;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
AVG_RATING
----------
       202
Floral Dress
         5

       203
Running Shoes
       4.5

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
AVG_RATING
----------

       205
Backpack
       4.5

       204
Analog Watch

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
AVG_RATING
----------
         4

       201
Cotton Shirt
       3.5
```

5.4 Customer-wise Total Orders

```
SQL> SELECT c.Customer_ID,
  2         c.First_Name || ' ' || c.Last_Name AS Customer_Name,
  3         COUNT(o.Order_ID) AS Total_Orders
  4  FROM Customer c
  5  JOIN Orders o ON c.Customer_ID = o.Customer_ID
  6  GROUP BY c.Customer_ID, c.First_Name, c.Last_Name
  7  ORDER BY Total_Orders DESC;

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
TOTAL_ORDERS
------------
        101
Rahul Sharma
           1

        102
Priya Ravi
           1

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
TOTAL_ORDERS
------------

        106
Divya Lakshmi
           1

        104
Anitha Suresh

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
TOTAL_ORDERS
------------
           1

        105
Arun Prakash
           1

        103

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
TOTAL_ORDERS
------------
Karthik Kumar
```

5.5 Customer-wise Total Purchase Amount

```
SQL> SELECT c.Customer_ID,
  2         c.First_Name || ' ' || c.Last_Name AS Customer_Name,
  3         SUM(p.Amount) AS Total_Purchase
  4  FROM Customer c
  5  JOIN Orders o ON c.Customer_ID = o.Customer_ID
  6  JOIN Payment p ON o.Order_ID = p.Order_ID
  7  WHERE p.Payment_Status = 'Successful'
  8  GROUP BY c.Customer_ID, c.First_Name, c.Last_Name
  9  ORDER BY Total_Purchase DESC;

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
TOTAL_PURCHASE
--------------
        104
Anitha Suresh
       5699.05

        102
Priya Ravi
        3398.3

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
TOTAL_PURCHASE
--------------

        103
Karthik Kumar
        2799.2

        106
Divya Lakshmi

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
TOTAL_PURCHASE
--------------
       2462.24

        105
Arun Prakash
        1619.1

        101
```

S

5.6 Payment Method Report

```
SQL> SELECT Payment_Method, COUNT(*) AS Total_Transactions, SUM(Amount) AS Total_Amount
  2  FROM Payment
  3  GROUP BY Payment_Method
  4  ORDER BY Payment_Method;

PAYMENT_METHOD                 TOTAL_TRANSACTIONS TOTAL_AMOUNT
------------------------------ ------------------ ------------
COD                                             1       1619.1
Card                                            2      9097.35
UPI                                             3      6561.44
```